

COMP 30640 Lab - 1

Getting to a terminal

On a Mac: open the Terminal.app and type 'bash'

On a BSD/Linux: open a terminal window and type 'bash'

On a Windows machine: You can either:

- Use the commandline that is built-in [cmd.exe]
- Download and use GitBash [<https://gitforwindows.org/>]
- Download and use PuTTY [<https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html>]
- Download and use MobaX-Term [<https://mobaxterm.mobatek.net/download.html>] - The free **HomeEdition** is good enough for this module

SSH to your account on Veridian

veridian.ucd.ie is the server that we will be using for access to a bash shell and demonstrating code.

[optional] If you have a Linux/BSD/Mac, then you may continue to use the bash shell that is built-in to your laptop.

[necessary] If you have a Windows machine, then using one of the tools mentioned previously, please log on to the server.

You should have received a password on your ucdconnect email

Your username is your student-ID [without the '@ucdconnect.ie' component]. For e.g., if your student-ID is: 123456, then you can ssh from a bash shell by typing:

```
ssh 123456@veridian.ucd.ie
```

When asked for a password, type in the password that you received in your email.

NOTE: When you log in for the first time, the system will ask you to change your password. Please ensure that you do so, and keep a strong password.

[Optional - if you don't want to use your password everytime]

Generate a private-public keypair using `ssh-keygen`

Copy your public key to veridian using `ssh-copy-id`

[**NOTE:** if you are on a windows machine, you may not have ssh-copy-id installed. Please refer to

<https://www.hanselman.com/blog/how-to-use-windows-10s-builtin-openssh-to-automatically-ssh-into-a-remote-linux-machine>

or

<https://chrisjhart.com/Windows-10-ssh-copy-id/>

to copy your ssh key to veridian]

Test that it worked. You should now be able to log into veridian by simply issuing the command:

```
ssh 123456@veridian.ucd.ie
```

Now onwards, you need not use your password to log into veridian as long as you're using the same machine/laptop

Navigating inside your bash

Now, we'll take a look at commands that you can use inside a shell to locate yourself and move around

`pwd` - Stands for **present-working-directory**. If you type this in, as soon as you log in and press enter, you should see something like

```
/home/<your-user-id>
```

For example, if you username is vivek, then your home directory would be:

```
/home/vivek
```

Using **man** to get help

Nearly every command and application in Linux will have a `man` (short for "manual") page, so finding them is as simple as typing `man` command to bring up a longer manual entry for the specified command.

For example:

```
$> man mv
```

will bring up the **mv (move)** manual.

Move up and down the man file with the arrow keys, and get back to the command prompt by pressing "q".

```
$> man man
```

will bring up the manual entry for the `man` command, which is a good place to start!

```
$>man intro
```

is especially useful - it displays the "Introduction to user commands" which is a well-written, fairly brief introduction to the Linux command line.

Some software developers prefer `info` to `man` (for instance, GNU developers), so if you find a very widely used command or app that doesn't have a man page, it's worth checking for an info page.

Virtually all commands understand the `-h` (or `--help`) option which will produce a short usage description of the command and its options, then exit back to the command prompt. Try `man -h` or `man --help` to see this in action.

man pages can be lengthy, so if you are looking for a specific option etc. it could be useful to look up some word using the syntax `/word` – and then hit the key `n` to move to the next occurrence. For example, try to find the option `-t` in the man page of `python`.

If you aren't sure which command or application you need to use, you can try searching the manual pages. Each manual page has a name and a short description.

If you know part of the command name, use the following command: `whatis -r <string>`. For example try with the following:
`$> whatis -r cpy`

To search the names or descriptions for `<string>` enter: `apropos -r <string>`.

For example:

```
$>apropos -r "copy files"
```

will list manual pages whose names or descriptions contain copy files.

File & Directory Commands

Veridian runs the Linux operating system. The Linux hierarchy is typical of Unix systems. For the moment you just need to know that the file system is a tree that starts at the root (represented with the symbol `/`). Note that if you are familiar with DOS/Windows the path delimiter is the forward slash (`/`) and not the backward slash (`\`). A path then looks like this in Linux: `/var/log/auth.log`. This leads to the file `auth.log` in the directory `log` in the directory `var` which is right inside the root of the file system.

- The tilde (`~`) symbol stands for your home directory. If you are vivek, then the tilde (`~`) stands for `/home/vivek`.
- `pwd`: The `pwd` command will allow you to know in which directory you're located (`pwd` stands for "print working directory"). Example: "`pwd`" in the Desktop directory will show `/home/vivek/Desktop`.
- `ls`: The `ls` (list) command will show you ('list') the files in your current directory. Used with certain options, you can see sizes of files, when files were made, and permissions of files. Example:

```
$>ls ~
```

will show you the files that are in your home directory. Check some of the options of ls.

- cd: The cd command will allow you to change directories. When you open a terminal you will be in your home directory. To move around the file system you will use cd.

Examples:

- To navigate into the root directory, use `cd /`
- To navigate to your home directory, use `cd` or `cd ~`
- To navigate up one directory level, use `cd ..`
- To navigate to the previous directory (or back), use `cd -`
- To navigate through multiple levels of directory at once, specify the full directory path that you want to go to. For example, use,
`$> cd /var/log`
to go directly to the /log subdirectory of /var/.

As another example,

```
$> cd ~/Desktop
```

will move you to the Desktop subdirectory inside your home directory.

- cp: The cp command will make a copy of a file for you. Example:

```
$> cp myfile foo
```

will make an exact copy of "myfile" and name it "foo", but the file called "myfile" will still be there. If you are copying a directory, you must use `cp -r directory foo` (copy recursively). (To understand what "recursively" means, think of it this way: to copy the directory and all its files and subdirectories and all their files and subdirectories of the subdirectories and all their files, and on and on, "recursively")

- mv: The mv command will move a file to a different location or will rename a file. Examples are as follows:

```
$> mv myfile foo
```

will rename the file called "myfile" to "foo".

```
$> mv foo ~/Desktop
```

will move the file called "foo" to your Desktop directory, but it will not rename it. You must specify a new file name to rename a file.

- rm: Use this command to remove or delete a file in your directory.

```
$> rm ~/Desktop/homework-1
```

will delete the file called "homework-1" from your Desktop directory

- rmdir: The rmdir command will delete an empty directory only. If there are files or other directories inside the directory, you must delete them first

- To delete a directory and all of its contents recursively, use `rm -r` instead.

- `mkdir`: The `mkdir` command will allow you to create directories. Example:

```
$> mkdir music
```

will create a directory called "music".

Exercises

1. Create a directory called "OS30640" inside your home directory
2. Create a sub-directory inside OS30640, called "lab-1"
3. Rename the directory "OS30640" to "lab-exercises"
4. List all the directories and sub-directories within your home directory
5. List all the directories and sub-directories within your home directory *ordered by* reverse chronological time of creation
6. Look up the manpage for `cat`, and display the contents of the file called "passwd", which is present in the `/etc` directory
7. Find the command that can be used to count characters and lines in files, and count the number of lines in the `/etc/passwd` file [*hint*: counting lines is the same as counting the number of newline characters in a file]
8. Copy the `passwd` file to the directory "lab-1" and check that the copied file has the *same number of bytes* as the original file using the same command used to count lines

System Status Commands

[specific to Linux systems such as Veridian]

There are several commands that can be used to find out the state of the system. The most frequently used among these are ones pertaining to disk storage and processes.

- `df`: The `df` command displays filesystem disk space usage for all mounted partitions. `df -h` is probably the most useful - it uses megabytes (M) and gigabytes (G) instead of blocks to report. (-h means "human-readable")
- `du`: The `du` command displays the disk usage for a directory. It can either display the space used for all subdirectories or the total for the directory you run it on. Two interesting options are `-s` ("summary") and `-h` ("human readable")

- **free**: The **free** command displays the amount of free and used memory in the system. **free -m** will give the information using megabytes, which is probably most useful for most laptops.
- **uname -a**: The **uname** command with the **-a** option prints all system information, such as name, kernel name & version, and a few other details. Most useful for checking which kernel you're using
- **lsb_release -a**: The **lsb_release** command with the **-a** option prints version information for the linux release that you're running